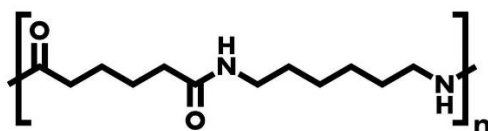


NYLON

Nylon is a synthetic polymer, it refers to a family of molecules called polyamides. It is also a thermoplastic which means that it can be moulded into different shapes at certain temperatures and solidifies after cooling, nylon softens at about 220°C.



nylon-6,6

Types & Preparation:

Types:

Nylon can be of different types, namely:

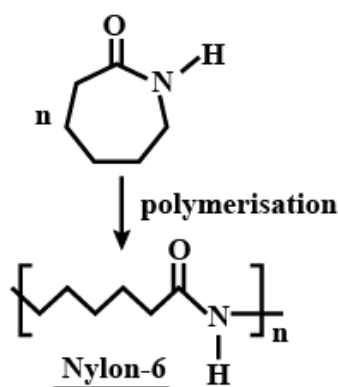
1. Nylon 6
2. Nylon 6,6
3. Nylon 5,10
4. Nylon 1,6

Preparation:

Nylon is named with respect to the monomers used to prepare them. Monomers react with other molecules to form polymers. The numbers indicate the number of carbon atoms present in the reactants or monomers. For example, in nylon 6,6, the two sixes indicate that each of the monomer involved in the production of nylon 6,6 has 6 carbon atoms each.

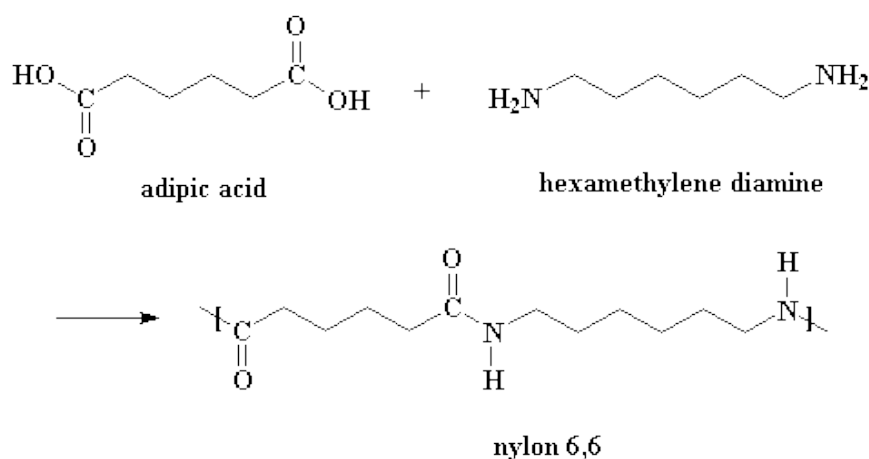
1. Nylon 6:

It is prepared by ring-opening polymerization of caprolactum $[(CH_2)_5C(O)NH]$. Caprolactum has 6 carbon atoms which is why the product is named Nylon 6. Ring-open polymerization refers to the process in which a cyclic monomer is polymerized to a linear polymer chain. Each cyclic monomer has a reactive center called terminal which is attacked to form a linear polymer chain.



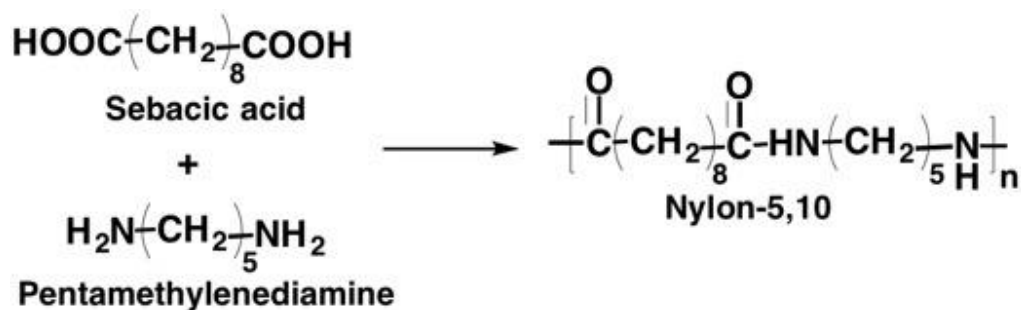
2. Nylon 6,6:

It is prepared by the condensation polymerization of the two monomers adipic acid and hexamethylene diamine at high temperature and pressure. Condensation polymerization refers to the step growth polymerization in which smaller polymer molecules react with each other to form larger molecules, while releasing biproducts such as water or methanol, which are referred to as condensates.



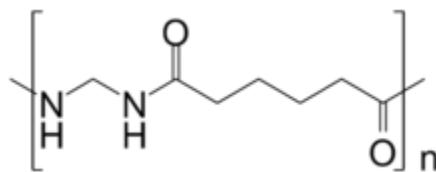
3. Nylon 5,10:

It is prepared by polymerization of sebacic acid and pentamethylene diamene.



4. Nylon 1,6:

It is formed by acid catalysis and not by condensation polymerization. Acid catalysis of adiponitrile, formaldehyde and water.



Nylon 1,6

Properties Of Nylon:

Physical:

1. Lustrous
2. Elasticity
3. Abrasion Resistance
4. Excellent tensile strength
5. Resilient and durable
6. Water resistant
7. Fast drying
8. Recyclable

Chemical:

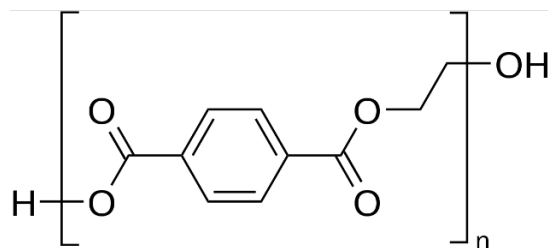
1. Acids: Nylons disintegrate when attacked by mineral acids. It is resilient to boiling organic acids, inert to formic and dilute acetic acids and is dissolved by concentrated formic acid.
2. Alkali: Inert to alkalis.
3. Organic Solvents: It is inert to most organic solvents but phenol metacresol and formic acid have the ability to disintegrate it.
4. Flammability: It burns rather slowly.
5. Bleaches: It can be harmed by chlorine and oxidizing solid bleaches, it is usually inert to oxidizing and reducing bleaches.

Uses Of Nylon:

1. It is used to make fishnets and fishing line.
2. Used to make clothing such as socks and stocking due to its elasticity.
3. Manufacture of machine parts and plastic fasteners.
4. Nylon 6,6 is used for making toothbrush bristles, parachutes, seatbelts, tubing hose etc.
5. Nylon 6 is used as insulation in circuit boards as well as machine guards.
6. Nylon 6 is often used in mechanical maintenance due to its stiffness and strength.

POLYETHYLENE TEREPHTHALATE (PET)

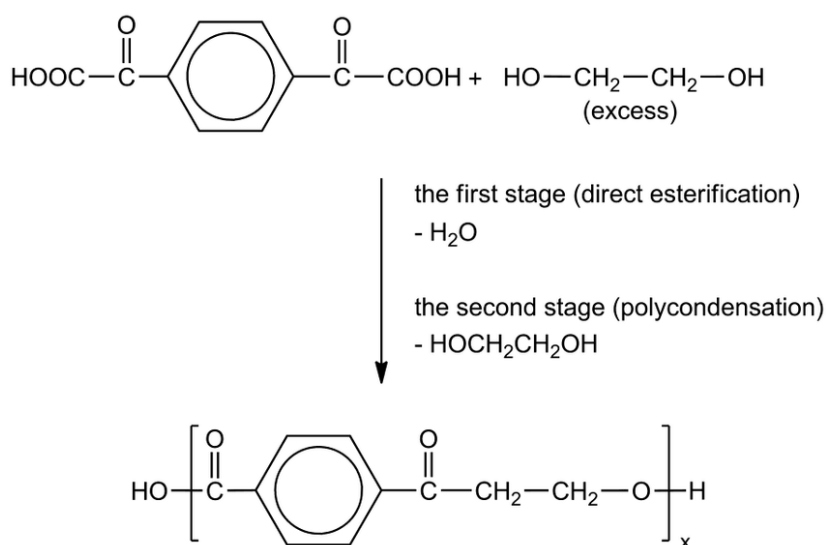
Polyethylene terephthalate (PET) is a type of polyester made from the monomers terephthalic acid and ethylene glycol. It produced in both solid and liquid forms, and commonly used to make beverage bottles and other packaging materials.



PET Molecule

Preparation:

PET or PETE plastic is made from the polymerization of a ethylene glycol and terephthalic acid. Ethylene glycol is a colourless liquid obtained from ethylene and terephthalic acid is a crystalline solid obtained from xylene.



Properties:

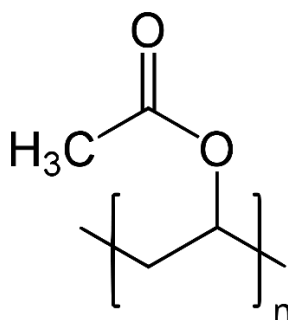
1. High rigidity and hardness
2. Flexibility
3. Very low moisture absorption
4. Chemical resistance against acids
5. Resistant to hydrolysis
6. Mechanical resistance
7. Good dimensional stability
8. Good adhesion and welding plastic
9. Lightweight
10. Clear

Uses:

1. To make water bottles, jars, carpets.
2. Ropes, automotive parts, fibre fill for winter jackets.
3. Microwavable trays, containers for cosmetic products, pharmaceutical products.
4. For manufacturing housing materials and clothes, sleeping bags and shopping bags.

POLYVINYL ACETATE (PVA)

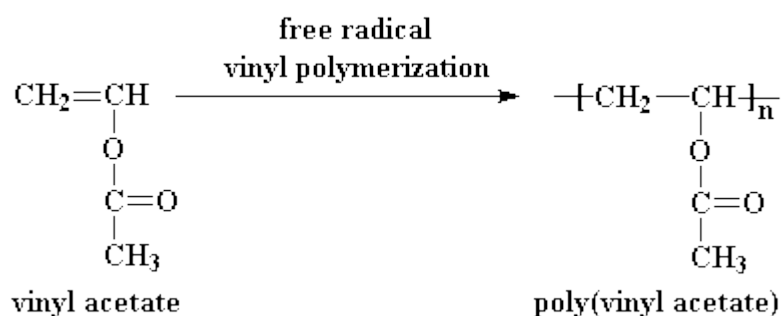
Polyvinyl acetate or PVA is a water-soluble synthetic polymer, commonly known as wood glue, PVA glue, white glue, carpenter's glue, school glue, or Elmer's glue in the US, is a widely available adhesive used for porous materials like wood, paper, and cloth. An aliphatic rubbery synthetic polymer with the formula $(C_4H_6O_2)_n$, it belongs to the polyvinyl ester family, with the general formula $-[RCOOCHCH_2]-$. It is a type of thermoplastic.



PVA Molecule

Preparation:

Polyvinyl acetate is prepared from the polymerization of vinyl acetate monomer (free-radical vinyl polymerization). It is done in the presence of a polymerization catalyst and then followed by partial hydrolysis.

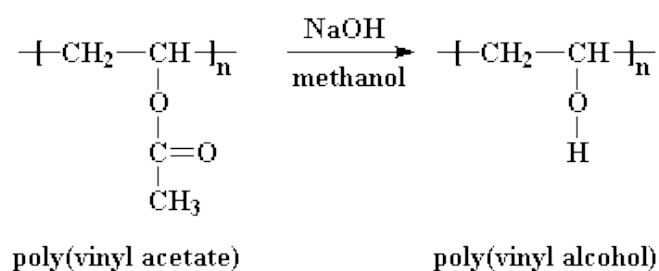


Properties:

1. It is non carcinogenic.
2. Weather resistant
3. Softens at about 30-45°C
4. Biocompatible and bioadhesive characteristics.
5. Biodegradable. A number of microorganisms can degrade polyvinyl acetate. Most commonly, damage is caused by filamentous fungi; however, algae, yeasts, lichens, and bacteria can also degrade polyvinyl acetate.
6. They can withstand water, grease, oil, and petroleum fuels.

Uses:

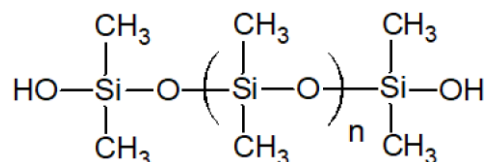
1. It is used as an emulsion to make sacks, bags and cartons.
2. It is used for making glass fibre-reinforced plastics to improve stress and antishrink properties.
3. PVA may be added to cement/concrete where it can improve the water-resistance properties.
4. PVA adhesives and copolymers are also used as hot-melt adhesives, sealants, in fabric finishing, plastic wood, and inks.
5. PVOH is produced by hydrolysis of PVA and the strong hydrogen bonding imparted by the –OH groups means that pure PVOH is water soluble.



6. PVOH is used for recycling paper and cardboard and is added to the slurry to improve the process.
7. Polyvinyl alcohol is widely used to strengthen textile yarn and papers, particularly to make the latter more resilient to oils and grease. It is also used in freshwater sports fishing.

SILICONE

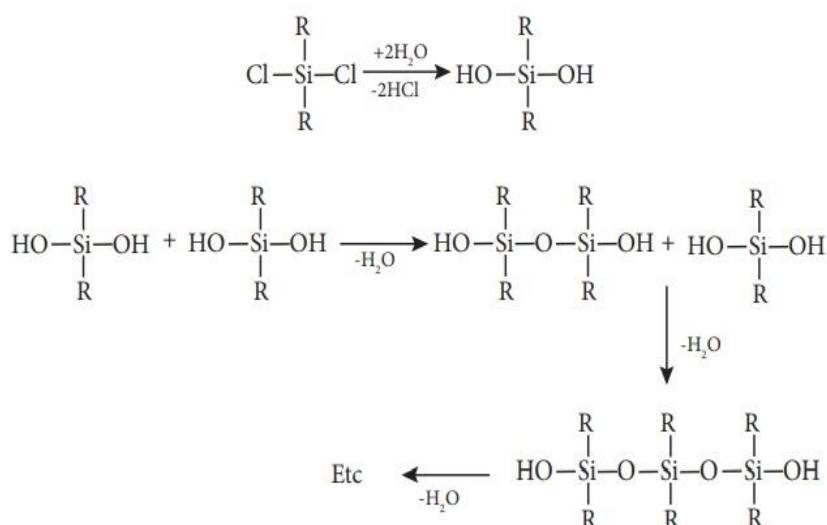
Silicone or polysiloxane is a polymer made of siloxane ($-\text{R}_2\text{Si}-\text{O}-\text{SiR}_2-$). They are typically colourless oils or rubber-like substances. It is a synthetic inorganic polymer whose main chain is made of polysiloxane, which is the repetition of silicon(Si) and oxygen(O) atoms.



Silicone Molecule

Preparation:

Hydrolysis of dimethyl dichlorosilane $[(\text{CH}_3)_2\text{SiCl}_2]$ followed by condensation polymerization which results in the formation of a straight chain silicone polymer.



Properties:

1. Silicones have low thermal conductivity and chemical reactivity.
2. Low toxicity.
3. Has high resistance to oxygen, ozone and UV light.
4. Has both electrical and conductive properties.
5. Insoluble in water and is hydrophobic.
6. It is highly elastic and compressible, due to the helical arrangement of the molecule and low intermolecular force.
7. It is thermally stable.
8. Silicones have low surface tension in the range of 20 to 25 mN/m.

Uses:

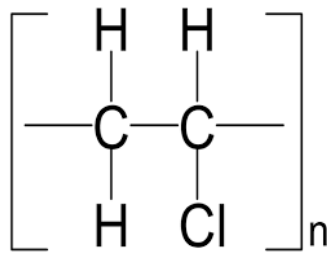
1. Silicones are widely used in personal care products. They are used to reduce the white residue and tacky feel of antiperspirants in deodorants. Silicones are used in cosmetics, shampoos and conditioners to retain the colour and lustre of the compound. Silicones have been said to offer better shine and skincare products can be made with stronger SPF.
2. Silicones are used extensively as a construction material for erecting commercial and residential buildings. They are said to prevent damage from moisture and bacteria build-up.
- 3.
4. Silicones are used to make keypads, keyboards and copier rollers. Many other components of computers, mobile electronics and home entertainment equipment make use of silicones. LED lighting technology is made possible by silicones. It has high thermal stability and excellent dielectric properties. So, it is used in a variety of electrical transmission applications.
5. Due to its high stress and temperature resistivity, they are used to seal and as protection for various parts of airplanes.
6. It is used as a sealant in the windows, doors, overhead bins, fuel tanks, engine gaskets, hydraulic switches, wings, wing edges, landing gear, electrical devices, vent ducts and even the black boxes.
7. Silicone-enhanced paints. These types of paints help to keep the exterior coatings of houses, bridges and railway cars flexible so they can withstand different temperature

cycles or weather conditions and especially prevent cracking. Silicone coatings are less likely to corrode.

8. Silicones are used as lubricants in many cases. Silicone greases are used to lubricate airsoft gun parts, bicycle chains and other mechanisms.
9. Silicone rubber an elastomer, is used in automotive applications, many cooking, baking, and food storage products, sportswear, and footwear, electronics, to home repair and hardware, etc.

POLYVINYL CHLORIDE (PVC)

Polyvinylchloride is a thermoplastic polymer. It is made by polymerizing vinyl chloride. It is made from resin which exists as white brittle granules.



PVC Molecule

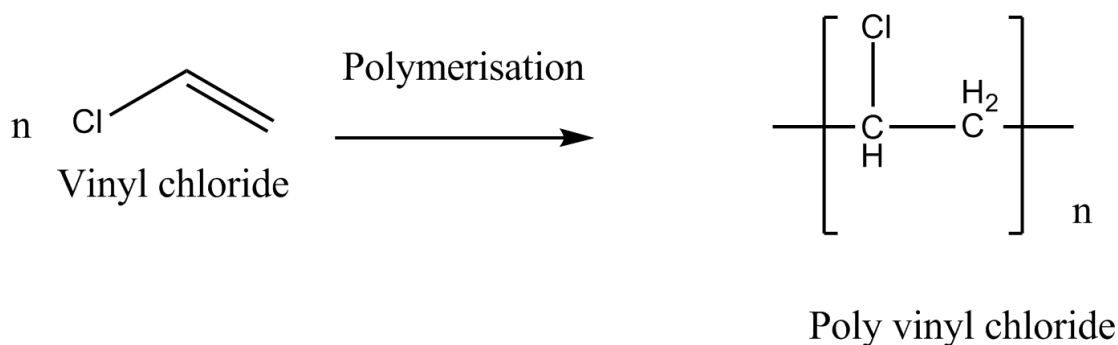
Types and Preparation:

It is of various types:

1. A rigid or unplasticized polymer (RPVC or UPVC)
2. Flexible plasticized PVC (PVC-P)
3. Chlorinated PVC
4. Molecular Oriented PVC (PVC-O)
5. Modified PVC (PVC-M)

Preparation:

PVC is formed as the result of polymerization of its monomer vinyl chloride. It is a chain growth polymerization reaction. The monomer is treated with peracid under pressure to obtain PVC.



Other Forms:

1. Plasticized or Flexible PVC: Flexible PVC is formed by the addition of compatible plasticizers to PVC which lower its crystalline nature. These plasticizers act like lubricants resulting in a much clearer and flexible plastic.
2. Unplasticized or Rigid PVC: Rigid PVC is a stiff and cost-effective plastic with high resistance to impact, water, weather, chemicals and corrosive environments it has little to no plasticizers added to it, also known as UPVC or PVC-U.
3. Chlorinated PVC: It is prepared by chlorination of PVC resin. High chlorine content imparts high durability, chemical stability and flame retardancy. CPVC can withstand a wider range of temperatures.
4. Molecular Oriented PVC or PVC-O: It is formed by reorganizing the amorphous structure of PVC-U into a layered structured. Bi-axially oriented PVC has enhanced physical characteristics (stiffness, fatigue resistance, lightweight, etc.).
5. Modified PVC or PVC-M: It is an alloy of PVC formed by addition of modifying agents, resulting in enhanced toughness and impact properties.

Properties:

1. PVC is lightweight, brittle, sturdy and an abrasion-resistant material.
2. Resistant to the action of all inorganic chemicals.
3. High dielectric strength and vapour barrier capacity.
4. It can withstand extreme climatic conditions, shock and is free from corrosion.
5. Extremely Durable.
6. PVC products are self-extinguishing due to high chlorine content.
7. PVC can be made more flexible and softer by adding plasticizers such as phthalate and can be bent as per requirement.
8. PVC is an intrinsic flame retardant.
9. Good tensile strength and is rigid by nature.
10. It is economical and is an affordable solution.
11. It demands less maintenance.
12. PVC is amorphous in nature, and hence it is easily combined with other chemicals/substances. Depending on additives used in manufacturing with PVC, many qualities can be imbued in products including anti-mist, different colours, elasticity, fire retarding, flexibility, impact resistance and microbe prevention.

Uses:

1. Plasticized PVC is used in flooring or (PVC-U) unplasticized PVC is used in making window frames.
2. It is used in making sewage pipes and other pipe applications where cost or vulnerability to corrosion limit the use of metal.
3. Used in construction fields for insulation on electrical wires or in flooring for hospitals, schools, homes, and other areas where a sterile environment is a priority.
4. It is used in various industries like building, electronics, electrical, automotive, medical and packaging.
5. PVC fabric is used in the manufacture of aprons shower curtains, raincoats, jackets and sports bags.
6. It is used in making garden hoses and imitation leather upholstery.
7. Used in making blood bags, tubing, wire and cable insulations, windshield system components etc.

